

JASMIN JAHAN PUSPO

✉ jasminjahanpuspo@gmail.com | 🌐 <https://jasminjahanpuspo.github.io/> | 📍 Sylhet, Bangladesh

EDUCATION

Master's in Computer Science and Engineering

Shahjalal University of Science & Technology

Jan. 2023 – Oct. 2025

CGPA: N/A

Bachelor's in Computer Science and Engineering

North East University Bangladesh

Jan. 2017 – Jun. 2021

CGPA: 3.54 / 4.0

RESEARCH INTERESTS

• Medical Imaging • Computer Vision • Machine Learning

PUBLICATIONS

Journal

- TransembleNet: Enhancing vector **mosquito species** classification through transfer learning-based ensemble model. *Plos One (Q1)*, 2025 🔗

Conference (First Author)

- Attention-Based CNN Architectures for Automated Binary Classification of Salmon **Fish Diseases**. *WIECON-ECE 2025* 🔗
- BengaliTaka: A Comparative Analysis of Transformer and CNNs on **Bangladeshi Currency** Recognition. *QPAIN 2025* 🔗
- SkinNet: An EnsembleNet Technique to Detect **Skin Cancer** Using Pre-Trained Models. *ECCE 2025* 🔗
- A Novel Approach to Classify **Breast Cancer** Using Transfer Learning. *ICCIT 2024* 🔗

RESEARCH EXPERIENCE

Research Collaborator

Elite Lab | Remote

Sep. 2025 - present

Dhaka, BD

- Working on image-data analysis using **Transformer** architectures and **XAI** (explainable AI) techniques.
- Collaborating with lab team to develop innovative research methodologies, improving efficiency and accuracy.

AI Researcher

SafeNet.AI | Remote

Jan. 2025 - Jun. 2025

Dhaka, BD

- Implemented **attention-based CNN** models and explored neural network architecture customization.

Research Intern

AL Lab | Remote

Sep. 2023 – Feb. 2024

Dhaka, BD

- Analyzed experimental results and authored the literature review for a study on **mosquito diseases**.
- Summarized from recent research papers and wrote a literature review catalog template

TECHNICAL SKILLS

Programming Language: Python, C

Deep Learning: CNNs, Transformers, Foundation Models

Framework : TensorFlow, Keras

Data Analysis: Pandas, Scikit-learn, OpenCV

Data Annotation : LabelImg, VGG Image Annotator

Tools: Git, JIRA

ACADEMIC THESIS

Graduate Thesis

Oct. 2024 - Oct. 2025

SUST | SUST Research Centre | PI: Dr. Mohammad Shahidur Rahman

Sylhet, BD

- Developed and curated a newly launched **Bangla banknote** dataset.
- Designed an explainable **hybrid CNN architecture** with multi-stage attention and vector feature fusion.

Undergraduate Thesis

Jun. 2020 - Jun. 2021

NEUB | PI: Muhammad Mahir Hasan Chowdhury

Sylhet, BD

- Conducted research on medical image classification using CNN models.
- Applied region-based preprocessing (**Mask R-CNN**) to enhance segmentation accuracy, and improved model focuses on **breast masses** by excluding pectoral muscles.

PROFESSIONAL EXPERIENCE

Content Writer

Oct. 2022 - May 2023

Ruskin Bright | Hybrid

Sylhet, BD

- Researched and created engaging **curriculum, book chapters**, and MCQ for various courses.
- Wrote sales content of 300-450 words by maintaining 100% quality and zero plagiarism.

TEACHING EXPERIENCE

Course Instructor | (3rd – 8th) grade

Jan. 2026 - present

AL Amanah Academy

Sylhet, BD

- Delivering lessons across all **STEM subjects**, incorporating practical demonstrations and interactive learning.

ICT Lecturer | (9th – 12th) grade

Feb. 2024

Women's Model College

Sylhet, BD

- Was responsible for daily 40-minute multimedia classes and delivered up to **four lectures** per day.
- Conducted classes covering technological topics, including theoretical coursework, and programming languages.

Academic Teacher | (3rd – 8th) grade

Sep. 2022 - Dec. 2022

Sylhet International School and College

Sylhet, BD

- Prepared lesson plans including laboratory class, lecture, exam, and homework.
- Graded assignments, tests, and lab work, providing feedback to help students improve academically.

Peer Tutor | (3rd – 12th) grade

May 2013 – Jun. 2024

- Designed lesson plans and materials to simplify concepts, enhancing student understanding and engagement.
- Assisted students with college admissions, supporting them in securing placements at reputable colleges.

DATASET COLLECTIONS

- Bengali Paper Currency 
- Bengali Sign Language 

STANDARDIZED TEST RESULTS

- IELTS Academic: Overall **6.0** (L: 6.0 | R: 6.5 | W: 5.5 | S: 6.0)

ACADEMIC SERVICE

- Conference Reviewer: **ICEF 2026**